Unit – 1

(2 Marks Questions)

1. What does HTML stand for?

2. Differentiate between HTML and CSS.

3. Explain the purpose of JavaScript in web development.

4. What is the role of a web browser in displaying a web page?

5. Name two commonly used web browsers.

6. Define responsive web design.

7. What is a CSS selector? Provide an example.

8. Explain the concept of a responsive image in web design.

9. Differentiate between margin and padding in CSS.

10. What is the box model in CSS?

11. What is a web server?

12. Describe the purpose of HTTP in web communication.

13. What is a URL? Provide an example.

14. Explain the difference between GET and POST HTTP methods.

15. What is a database in the context of web development?

16. Define the term "API" in web development.

17. What is a session in the context of web applications?

18. Explain the purpose of cookies in web development.

19. Differentiate between front-end and back-end development.

20. What is version control in the context of web development?

5-mark questions

1. Explain the concept of Responsive Web Design (RWD) and its importance in modern web development. Provide examples of techniques used to achieve responsiveness in web design.

2. Compare and contrast HTTP and HTTPS protocols. Discuss the significance of using HTTPS for securing web communication and the steps involved in obtaining an SSL/TLS certificate.

3. Describe the purpose and usage of JavaScript Promises in asynchronous programming. Provide an example of how Promises can be used to manage asynchronous tasks in a web application.

4. Explain the role of the Model-View-Controller (MVC) architectural pattern in web development. Illustrate how this pattern promotes separation of concerns and discuss its benefits.

5. Discuss the differences between client-side scripting and server-side scripting in web development. Provide examples of programming languages commonly used for each approach and situations where one might be more appropriate than the other.

6. Elaborate on the concept of Cross-Origin Resource Sharing (CORS) in web development. Explain why CORS is important, how it prevents unauthorized access, and methods to configure CORS policies on a web server.

7. Describe the purpose and usage of SQL injection attacks. Explain how web developers can prevent SQL injection vulnerabilities in their applications and provide examples of best practices.

8. Explain the concept of Single Page Applications (SPAs) and how they differ from traditional multi-page websites. Discuss the advantages and challenges of building SPAs, along with technologies commonly used to develop them.

9. Discuss the role of caching in web development. Explain the different types of caching (browser caching, server-side caching, content delivery network caching) and how they improve website performance.

10. Elaborate on the importance of accessibility in web development. Describe the Web Content Accessibility Guidelines (WCAG) and provide examples of design and development practices that make websites more inclusive for users with disabilities.

Remember, for each of these questions, it's important to provide clear and concise explanations, supported by relevant examples where applicable.

Unit – 2

(2 Marks Questions)

1. What is frontend development?
2. Name two programming languages commonly used in frontend development.
3. Explain the role of HTML in frontend development.
4. What is the purpose of CSS in frontend development?
5. Define the term "DOM" in the context of frontend development.
6. What is responsive web design, and why is it important in frontend development?
7. Explain the difference between inline, internal, and external CSS.
8. What is a CSS framework? Provide an example.
9. Define the concept of "box model" in CSS.
10. What is a media query in responsive design?
11. Explain the role of JavaScript in frontend development.
12. What is an event handler in JavaScript?
13. Differentiate between "let," "const," and "var" in JavaScript variable declarations.
14. What is AJAX, and how is it used in frontend development?
15. Define the term "SPA" in the context of frontend development.
16. What is the purpose of a frontend build tool like Webpack?
17. Explain the role of a frontend framework like React or Vue.js.
18. What are the benefits of using SVG (Scalable Vector Graphics) in frontend development?
19. Define "cross-browser compatibility" and why it's important in frontend development.
20. What is the purpose of browser developer tools in frontend development?

Unit – 3

(2 Marks Questions)

1. What is React.js?
2. Explain the concept of "virtual DOM" in React.
3. What is JSX in the context of React?
4. Define a React component.
5. Differentiate between functional components and class components in React.
6. What is state in React?
7. Explain the purpose of props in React components.
8. What is the significance of the "render" method in React components?
9. Define React hooks. Provide an example of a hook.
10. Explain the role of the "useState" hook in managing state in a functional component.
11. What is the purpose of the "useEffect" hook in React?
12. Define React Router.
13. Explain the purpose of the "Link" component in React Router.
14. What is conditional rendering in React?
15. Explain the concept of "unidirectional data flow" in React.
16. What are React props validation?
17. Define React context.
18. Explain the concept of "controlled components" in React forms.
19. What is the role of the "key" attribute in React lists?
20. Explain the term "reconciliation" as it relates to React's rendering process.

Unit – 4

(2 Marks Questions)

1. What is Java EE (Enterprise Edition) for web development?

2. Explain the role of Servlets in Java web development.

3. Define JSP (JavaServer Pages) in the context of web development.

4. Differentiate between Servlets and JSP.

5. What is the purpose of a JavaBean in web development?

6. Explain the concept of session management in Java web applications.

7. Define the role of JDBC (Java Database Connectivity) in web development.

8. What is the significance of the web.xml file in Java web applications?

9. Explain the MVC (Model-View-Controller) architecture in the context of Java web development.

10. What is the purpose of the "request" and "response" objects in Java web applications?

11. Define the term "container" in Java EE web development.

12. Explain the difference between "GET" and "POST" HTTP methods in the context of Servlets.

13. What is a JSP tag library? Provide an example.

14. Define the purpose of a filter in a Java web application.

15. Explain the role of EJB (Enterprise JavaBeans) in Java EE web development.

16. What is a servlet container? Name one popular servlet container.

17. Define the concept of "dependency injection" in Java web development.

18. Explain the purpose of JNDI (Java Naming and Directory Interface) in Java web applications.

19. What is a JAR file in Java web development?

20. Define the term "JavaServer Faces (JSF)" and its use in web development.

Unit – 5

(2 Marks Questions)

Databases:

1. What is a database management system (DBMS)?
2. Differentiate between SQL and NoSQL databases.
3. Explain the primary purpose of a primary key in a database table.
4. Define database normalization.
5. What is an index in a database, and why is it important?
6. Explain the difference between a database and a database schema.
7. Define ACID properties in the context of database transactions.
8. What is a foreign key in a database table?
9. Explain the concept of database replication.
10. Define a stored procedure in the context of databases.

Deployment:

1. What does "deployment" mean in software development?
2. Explain the difference between development, testing, and production environments.
3. What is version control, and how does it aid in deployment?
4. Define continuous integration (CI) and continuous deployment (CD).
5. Explain the purpose of a reverse proxy server in deployment.
6. What is Docker, and how does it facilitate application deployment?
7. Define load balancing and its role in deploying web applications.
8. Explain the difference between horizontal and vertical scaling in deployment.
9. What is a Content Delivery Network (CDN), and how is it used in deployment?
10. Define "zero-downtime deployment" and why it's important.